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An Exploratory Study of the Relationships between an Assessment of Problem Solving Style and Creative Problem Solving

Scott G. Isaksen
The Creative Problem Solving Group, Inc.

David Geuens
VLEKHO Hogeschool for Science and Art, USA

The purpose of this study was to examine the relationships between VIEW: An Assessment of Problem Solving Style (VIEW; Selby, Treffinger, Isaksen & Lauer, 2004; Treffinger & Selby, 2004), and the preference for learning and using of the specific tools, guidelines, stages and components of the current version of creative problem solving (Isaksen & Treffinger, 2004; Treffinger, Isaksen, & Stead-Dorval, 2006). This study, conducted by Geuens (2006), can be considered a part of a larger program of research called the Cognitive Styles Project, so the previous work done within the program will be summarized to provide an understanding of the broader context for the research.

Significant relationships between all three dimensions of VIEW and various creative problem solving (CPS) tools, guidelines and process stages were found. Findings on the Orientation to Change dimension from VIEW were similar to previous findings using the Kirton Adaption-Innovation Inventory (KAI). New insights for the Manner of Processing dimension of VIEW showing, for example, that internals do favor learning Brainwriting. New insights were also found for the Ways of Deciding dimension of VIEW, showing that Task-Oriented deciders reported many more significant relationships for both learning and applying CPS.

In general, even though this was a very exploratory study, there was support for the use of VIEW in improving our understanding of style differences for learning and applying CPS.

COGNITIVE STYLES PROJECT

The Cognitive Styles Project (CSP) is a program of research that fits within a larger creativity research initiative described as ecological (Harrington, 1990; Isaksen, Puccio & Treffinger, 1993). The main idea of an ecological framework is to assert that the most complete picturing of creativity stems from considering all aspects of the creativity system. This includes examining the contingencies surrounding the creative person, the creative process, the outcomes of creativity, as well as the environment or climate that supports creativity. The CSP was initiated at the Center for Studies in Creativity, Buffalo NY, and was launched in the early 1980's to study the links bet-

Correspondence concerning this article should be addressed to Scott G. Isaksen, Creativity Research Unit, the Creative Problem Solving Group, Inc., P.O. Box 648, 6 Grand View Tail, Orchard Park, New York 14127, USA. E-mail: sgiaway@cpsb.com

ween person and process. The origin of the CSP stems from an earlier research initiative, the Creative Studies Project, also conducted in Buffalo.

The Creative Studies Project was based on the work of Osborn (1953) and Parnes (1966). The Osborn-Parnes approach to CPS was examined experimentally to see if training in this approach would positively impact the creative ability of the participants. Some of the earliest studies conducted by Parnes and his associates evaluated the effects of specific CPS programs and methods (Meadow & Parnes, 1959; Meadow, Parnes & Reese, 1959; Parnes, 1961, 1963; Parnes & Meadow, 1959, 1960), however there was a general lack of experimental evidence of the impact of an eclectic and comprehensive training effort on a variety of creative abilities.

The Creative Studies Project began with a pilot program at the State University College at Buffalo in 1969, and included a four-semester series of creative studies courses for the experimental group. This two-year experimental project provided enough empirical support for the undergraduate coursework to enable the college to approve the upgrading of these courses to regular credit-bearing elective status in 1972 (Khatena & Parnes, 1974; Noller & Parnes, 1972; Parnes, 1987; Parnes & Noller, 1972 a & b; Parnes & Noller, 1973; Reese, Treffinger, Parnes & Kaltsounis, 1976). This instructional program has become well established in the research activities of other scholars (Basadur, Graen & Green, 1982; Buijs & Nauta, 1991; Cramond, Martin & Shaw, 1990; De Schryver, 1992; Geschka, 1993; Puccio, Firestien, Coyle, & Masucci, 2006; Rose & Lin, 1984; Torrance, 1972, 1986, 1987; Shack, 1993).

Although the two-year program was very successful for the experimental subjects who remained for the entire program, there were some subjects who chose to drop out of the program. The experimental and control subjects who continued the program were initially comparable on nearly all the personality assessments conducted. There were some interesting findings, however, regarding those experimental and control group members who dropped out. They possessed characteristics such as: more directed toward deviancy or culturally disapproved behavior, in closer contact with their primary processes, freer, more impulsive, more likely to drop out of college, less responsible and more anxious (Parnes, 1987). Dropouts seemed to be more interested in artistic forms of creativity and withdrew because of their disappointment in the nature of the course. The implications and more extensive description of the findings of the dropouts are reported extensively in Parnes & Noller (1973) and Parnes (1987). This link to individual differences formed the basis for the CSP.

The purpose of the CSP was to better understand the individual differences associated with learning and applying CPS. The aim was stated as improving our understanding of what works for whom, and under what circumstances (Isaksen, 1987). Those who have collaborated on this project have produced more than 30 master's theses, doctoral dissertations, and scholarly publications. Although this project was initiated at the Center for Studies in Creativity, work has continued at the Creative Problem Solving Group and its Creativity Research Unit, the Center for Creative Learning, and by other interested colleagues and students.

Definitions of Style

There is a diversity of definitions, theories, constructs, and assessments of cognitive styles (Vernon, 1973). Despite the lack of clarity, a few trends are discernible. For example, nearly all these constructs of style seem to differentiate style from ability. Kogan (1976) indicated that the general ability domain focuses on level of perfor-

mance, whereas the cognitive style domain deals with the manner in which individuals acquire, store, retrieve, and transform information. Despite the apparent and broad interest, there are also those who do not place much value on the concept of cognitive styles given the apparent confusion between ability and modality and a lack of robust programs of research (Miller, 1987; Tiedemann, 1989).

The study of cognitive styles has also been seen as a subset of the discipline of cognitive psychology. Hayes (1978) indicated that cognitive style was concerned with a variety of mental processes including memory, learning, comprehending language, problem solving, and creativity. He also asserted that cognitive style existed on the borderline between intellectual function and personality. Willerman (1979) referred to cognitive styles as filter mechanisms that function to sort what and how we perceive and process information.

Within the field of cognitive psychology, there is considerable interest and effort being focused on the issue of cognitive styles. Goldstein and Blackman (1978) discussed the nature of cognitive styles and explained that it is a hypothetical construct designed to explain the mediation between stimulus and response in a manner that describes characteristic ways in which individuals conceptually organize the environment. Martinsen and Kaufmann (1999) have identified five main issues associated with cognitive styles and creativity. They point out that: most researchers have made a distinction between cognitive styles and abilities, cognitive styles can be placed at the intersection between personality and cognition, styles may be more strongly related to higher-order strategies rather than task-specific ones, most style constructs have been defined as being bipolar, and theories of cognitive style need to be related to more general theories of cognition and personality.

Guilford (1980) indicated that what some call cognitive styles may be called cognitive controls, cognitive attitudes or cognitive system principles. Messick (1984) added cognitive preferences, structural properties of the cognitive system, preferred or habitual decision-making strategies, ingrained strategies of learning and knowledge acquisition, and cognitive consequences of personality trends to the ever increasing list of cognitive-style conceptualizations. He reviewed the variety of definitions and proposed that all the alternative conceptions imply consistent individual differences, suggest deeper roots in personality structure, and tend to be pervasive.

Cognitive styles have also been linked to cognitive strategies (Kaufmann, 1995). Styles seem to be more spontaneously applied without conscious deliberation, whereas strategies seem to be more a matter of choice and training. The areas of learning strategies and meta-cognition are also intimately linked to these two concepts (Forrest-Pressley, MacKinnon & Waller, 1985; Weinstein & Mayer, 1986; Wittrock, 1986). Messick (1984) pointed out that forming and choosing cognitive strategies may be subject to change through conditions of learning. He asserted that individuals are able to learn a variety of strategies, some of which may be more consonant with their preferred style. He also asserted that individuals might be able to make a shift to less congenial strategies that are more effective for particular tasks.

The broader concept of style has been linked with learning style (Dunn & Dunn, 1978), psychological type (Myers & McCauley, 1985), cognitive style (Kirton, 1978, 1994) and more recently to problem-solving style (Selby, Treffinger, Isaksen & Lauer, 2004). Selby, Treffinger, & Isaksen (2007a & b) defined problem-solving styles as consistent individual differences in the ways people prefer to plan and carry

out generating and focusing activities, in order to gain clarity, produce ideas, and prepare for action.

Results from the Cognitive Styles Project

One of the initial thrusts of the CSP was aimed at an improved understanding of various approaches to understanding of individual differences by examining numerous style assessments. Wittig (1985) conducted the initial study by examining the relationship between children's learning style and varying levels of divergent thinking ability. She used Dunn & Dunn's Learning Style Inventory (Dunn & Dunn, 1978) to assess the style preferences of third-grade children and found various significant differences in those students with high and low divergent thinking ability. Her study was replicated and extended by McEwen (1986) who found similar results with older children.

Corbett-Whitier (1986) also used Dunn and Dunn's Learning Style Inventory (LSI) as a measure of style and examined its relationship to the various factors of the Torrance Tests of Creative Thinking (TTCT; Torrance, 1974). She also found a number of significant relationships between the LSI and the TTCT.

The next major phase of the project was to examine the available instrumentation to see which assessment approaches would be most fruitful for use in the instruction-oriented research program. The project then involved the Myers-Briggs Type Indicator (MBTI; Myers & McCaully, 1985), the Kirton Adaption-Innovation Inventory (Kirton, 1976), and the Style Delineator (Gregorc, 1982).

Joniak and Isaksen (1988) examined the psychometric properties of and relationships between the KAI and Gregorc's Style Delineator. The KAI performed consistently with previous research regarding its factor structure and internal consistency, but the Style Delineator was dropped from the project due to its inadequate psychometric properties (O'Brien, 1990; Sewall, 1986).

Level and Style of Creativity

Another major thrust of the CSP was the examination of the conceptual separation of creative level or ability from modality or preferred style. A number of publications have summarized the progress and potential of this level-style distinction (Isaksen, 2004; Isaksen & Dorval, 1993; Kaufmann, 2004).

Isaksen and Puccio (1988) studied the relationship between KAI, a measure of style, and the TTCT, a measure of creative level. They reported small, positive, and signify-cant correlations between the abilities of fluency, flexibility and originality of the TTCT and an innovative style of creativity on the KAI.

Teft (1990) conducted a follow-up to this study. She used factor-analysis to examine the relationship among the TTCT, KAI, and MBTI. Her results confirmed Kirton's assertion and Goldsmith's (1987) findings of an orthogonal relationship between creative level and creative style. She reported that the TTCT and the KAI loaded on two separate factors (level and style, respectively). However, she also found relationships between the KAI and the "Creativity Index" and two of the dimensions of the MBTI. She concluded that the KAI was a sound measure of creativity style and that the assessments used to measure creative level may have been contaminated with style variables.

One implication of her findings concerns the use of the MBTI, a measure of style, as a means for determining level of creative performance. In her study, Teft reported a strong correlation between the Creativity Index (CI) of the MBTI and the style fac-

tor, as well as a lack of significant correlation between the CI and the level factor. This suggests that the CI is more a measure of style than level. However, as Teft pointed out, Gough (1981) developed the CI to predict level of creative talent. Theoretically, the MBTI is a measure of style and should not be able to predict level. Gough's use of a style measure to predict level of creative talent demonstrates the conceptual confusion that exists in creativity research and practice. It also suggests that the level-style distinction may be helpful to researchers interested in assessing creative performance by providing conceptual clarity to creativity constructs and forcing the development of better and more conceptually sound measures of level and style.

The CSP broadened its focus by structuring an examination of the concept of mental imagery as a mode of cognitive processing. Dorval (1990) used the level-style distinction to organize the measures used to assess level and style of both creativity and imagery. He used the Revised Minnesota Paper Form Board Test (MPFB) developed by Likert and Quasha (1941) to measure visual imagery ability and the Individual Differences Questionnaire (IDQ) developed by Paivio (1971) to measure preference for imaginal and verbal modes of cognitive processing.

Similar to previous results (Isaksen & Puccio, 1988), Dorval (1990) reported small but significant relationships between the measures of level and style of creativity. However, he found no significant relationships between the MPFB and the imaginal or verbal scales of the IDQ. The strongest findings concerned the relationship between creativity style and preference for imagery. Those with an innovative style on the KAI reported stronger preferences for both imagery and verbal modes of cognitive processing. These results suggested the IDQ was a measure of style or preference for mode of cognitive processing, while the MPFB test was a measure of level of visual imagery ability. Finding no significant relationships between assessments of level and style of imagery supported Dorval's assertion that the level-style distinction found in the general creativity domain was also present for the concept of imagery.

This line of research continued with work conducted by Isaksen, Dorval and Kaufmann (1992). They focused specifically on the style relationships between creativity and imagery reported by Dorval (1990). They found that individuals with an innovative style of creativity had a higher general level of preference for conscious symbolic processing (both verbal and imaginal in nature). Those with an innovative preference on the KAI had a stronger preference for the imaginal mode of symbolic processing over the verbal processing mode. These findings again suggested there are individual differences in how people prefer to use cognitive processing in their approaches to creativity and that the level-style distinction may be a productive approach to understanding creativity and related concepts.

Other results

Further work within the CSP included pursuing a deeper understanding of style and exploring the applications of the KAI. For example, Selby (1991) examined the psychometric suitability of the KAI for students of middle school age. Selby, Treffinger, Isaksen and Powers (1993) published the results from this dissertation and indicated that using the KAI with younger students was not recommended as some of the inventory items would need adjustment before widespread use with this age group.

Isaksen and Kaufmann (1990) studied the relationship between cognitive style and perceptions of the climate for creativity. Using the KAI as a measure of style and the Situational Outlook Questionnaire (SOQ; Isaksen & Ekvall, 2006) as a measure of psychological climate, they found few significant relationships between the two constructs. Follow up studies (Grivas, 1996; Isaksen & Lauer, 1999) found similar results on a quantitative level, but also found meaningful qualitative differences between innovators and adaptors and their perceptions regarding what would help or hinder their creativity at work.

Kaufmann, Isaksen and Lauer (1996) used the KAI in a study to explore the Glass Ceiling effect in organizations. They studied women in managerial roles and found that females in these roles had a stronger preference for an innovative style of createvity and problem solving. In discussing their findings, they suggested that females pursuing managerial positions needed to utilize these preferences to break through the glass ceiling within many organizations.

In another study with organizational implications, Isaksen, Babij and Lauer (2003) studied the relationship between the KAI and the Leadership Practices Inventory (LPI). They found significant correlations between KAI and two of the six dimensions of the LPI, a measure designed to assess behaviors associated with extraordinary leadership. Both the Challenging the Process and Inspiring a Shared Vision dimensions of the LPI were found to relate to a more innovative style preference on the KAI.

The next major phase of the Cognitive Styles Project focused on identifying the key dimensions of style that would help provide insight to those learning and applying CPS. Houtz, et al. (2003) examined the relationship between the KAI and the Basadur CPS Profile (Basadur, 1991), an assessment with direct application to learning and applying CPS. They reported that Basadur's generator scores were significant and positively correlated to all scores on the KAI, illustrating a key relationship between the KAI and at least one aspect of CPS.

Isaksen, Lauer and Wilson (2003) studied the relationship between the KAI and the MBTI. They summarized the findings from previous research and used a larger sample in their study. They found that the KAI was significant and positively correlated with the Intuitive perceptual preference from the MBTI. No significant correlations were found between the KAI and preferences for Introversion – Extroversion or Thinking – Feeling dimensions on the MBTI.

In a more current phase of the CSP, Selby, Treffinger, and Isaksen (2007a & b) have developed a measure called VIEW to assess problem-solving styles. They were influenced by their practical experiences in education and business as well as Kirton's theory and KAI, Jungian theory and the MBTI, Dunn and Dunn's approach to understanding and measuring learning styles, and the work of Martinsen and Kaufmann (1999) on assimilators and explorers from a Piagetian perspective. VIEW includes three independent dimensions of problem-solving style: Explorer-Developer Orientation to Change, Internal-External Manner of Processing, and Task-Person Ways of Deciding. These will be defined and described in more detail later.

Style and CPS

Since one of the main aims of the CSP was to understand individual differences in learning and applying CPS, many of the studies examine the general relationship of style to various aspects of CPS. DeLuca (1981) conducted an initial study within the

project. She examined the impact of a special education program including CPS and how differences in style affected the results. Zilewicz (1986) found that students with different styles reported strong differences in how they learned and applied CPS methods and tools. He used Gregorc's (1982) measure of style and found that undergraduate students in creative studies classes, whose scores were different on the Style Delineator, showed clear differences in the way they perceived and ordered information, generated and evaluated ideas, worked with groups, and carried out plans of action. In short, he found that individuals with different cognitive styles reported different strengths and weaknesses. When students had similar cognitive styles, they demonstrated similar strengths and weaknesses on various aspects of the CPS framework.

Puccio (1987) found that scores on the KAI were able to discriminate the fluency and originality of problem statements generated by students when solving a real task provided from the railroad industry. In examining the effect of style and fluency on the originality of problem statements, he found that fluency accounted for most of the variance in the predicted behavior. Although Puccio's results did not support Kirton's orthogonality assertion, he suggested that further investigation of the concepts was necessary to bring additional clarity to the level-style question.

Murdock, Isaksen and Lauer (1993) examined the stability of cognitive style preferences by giving the KAI to students before and after a full-semester course in CPS. They found that student's cognitive style preferences were not affected by a full semester of deliberate training in creative approaches to problem solving.

A series of studies examined the relationship between style and preferences for learning and the application of various CPS tools. Hurley (1993) found quantitative and qualitative relationships between style of creativity as measured by the KAI and the use and application of specific CPS tools and techniques. Key quantitative differences showed that adaptors enjoyed idea systems and deliberate converging or focusing tools and strategies. Innovators preferred more numerous divergent idea-generating tools and more primary convergent tools. Qualitatively, adaptors enjoyed practicing the strategies in order to feel comfortable. On the other hand, once those with an innovator preference had an insight, they immediately wanted to converge or focus, and then implement the idea. Schoonover (1996a & b) replicated these findings and added insight into which specific tools were preferred and used more frequently by both adaptors and innovators.

Wheeler (1995) used the Productivity Environmental Preference Survey (PEPS; Price, Dunn & Dunn, 1991), the Strong-Campbell Interest Inventory (Strong & Campbell, 1974), and a survey designed to examine behavioral preferences for stages and components of CPS. He reported numerous significant correlations between the PEPS and the Strong-Campbell Interest Inventory and self-reported preferences associated with various stages and components of CPS.

Although specific CPS tools are likely to be style neutral, there is evidence that people of varying style orientations have distinctly different preferences for particular kinds of tools, phases within stages, and specific stages of CPS (Basadur, Graen & Wakabayashi, 1990; Rickards & Puccio, 1992).

The majority of the recent studies within the CSP used the KAI as the measure of style. The KAI is based on a single theoretical continuum called adaption-innovation. In addition, even though a main theoretical thrust of KAI theory is the level-style distinction, one study raised serious limitations to the actual use of the measure.

Chimento (2000) studied the perceptions of 188 people regarding the descriptions of the adaptor and the innovator. She found that regardless of an individual's score on the KAI, there was a clear and ubiquitous finding that people find the description of the innovator to be more creative.

These empirical findings were consistent with feedback and experience from various educational and business applications of the KAI. If one of the key theoretical positions underpinning the KAI is the conceptual separation of level from style, then these findings raise serious implications for both the nomenclature of the measure and its practical use as well.

Secondly, the KAI contains one main dimension with which to assess and understand style differences. Our research (Isaksen, Lauer, & Wilson, 2003) indicated that the conceptual overlap between the KAI and MBTI yielded three main stylistic dimensions. The first appeared to blend Intuition, Perceiving and the Innovator orienttation on the one side; and Sensing, Judging, and Adaption orientation on the other. The study produced no relationship with the two remaining dimensions of the MBTI (Introversion-Extraversion and Thinking-Feeling).

As a result of these empirical findings and our applied work, VIEW was developed to assess three independent dimensions of problem-solving style with careful attention to use language that was not style-biased. Since most previous research used other measures, it was important to initiate an exploratory study to examine VIEW's usefulness in understanding the empirical relationship to learning and applying CPS. This represents an initial study aimed at accomplishing this goal.

METHOD

Invitations to participate in the study were sent to 134 people. Each had completed an Igniting Creative Potential (ICP) course. This three-day course introduces each participant to 17 creative problem-solving tools, eight guidelines (four generating and four focusing), as well as to eight specific stages of the creative problem solving process. Trainers of all courses utilized the same design. The ICP course is based on the current version of CPS called CPS version 6.1 (Isaksen & Treffinger, 2004; Treffinger & Isaksen, 2005). A variety of instructional materials are used in the course (Isaksen & Tidd, 2006: Isaksen, Dorval & Treffinger, 2000; Treffinger, Isaksen & Stead-Dorval, 2006).

The sample consisted of two main groups of participants. Eighty-nine subjects had completed the ICP course provided by the Creative Problem Solving Group (CPSB) and forty-five had completed the course through training provided by the US Department of Defense (DOD). Completed surveys were obtained from 81 subjects, providing a response rate of 60%. Fifty-one of the 89 from CPSB, and 30 of the 45 from the USDOD completed the surveys. The CPSB participants were primarily managers within large US organizations who had some responsibility for innovation, new product development, and facilitation of meetings. They had a variety of other training experiences with other change methods and held various change and improvement-oriented positions within their organizations. These individuals were mainly North American. The DOD participants were drawn from the US Army Materiel Command involved in a Lean-Six Sigma change effort. The sample consisted of 45 males with a mean age of 45, and 36 females with a mean age of 39. These individuals were also mainly drawn from North America.

Participants had completed VIEW during their training, so these data were readily available. VIEW assesses three independent dimensions of problem-solving style. Orientation to Change (OC) addresses the questions: "How do I prefer to deal with boundaries and parameters?" "How do I feel about and react to structure?" and "How do I prefer to respond to novel challenges?" Scores below the mean on this dimension indicate the Explorer style. In ordinary use, an "explorer" is an individual who thrives on venturing in uncharted directions, seeks to break new ground, and follow adventurous or promising new possibilities wherever they may lead. Scores above the mean on the OC scale indicate the Developer style. In ordinary use, a "developer" is an individual who brings tasks to fulfillment, who begins with the basic elements or ingredients and then organizes, synthesizes, refines, and enhances them, forming or shaping them into a more complete, functional, useful condition or outcome.

The second scale within VIEW is called Manner of Processing (MP) and addresses the questions: "How do I prefer to manage information and its flow when problem solving?" "When do I share my thinking?" and "Does interacting with others build or spend energy?" Scores below the mean indicate a preference for an "External" style of processing. Individuals who exhibit a well-developed preference for this style draw their energy from interaction with others, discussing possibilities, and building from the ideas of others. Scores above the mean reflect a preference for an "Internal" style of processing. Those with a well-developed Internal style look first reflectively to their own inner resources and draw energy from their reflection.

The third scale within VIEW is called Ways of Deciding (WD) and addresses such questions as: "What factors get first priority when I focus or decide?" "Where do I start?" and "How do I make trade-offs?" Scores on this scale indicate whether one's primary focus in decision-making is on "People" or "Task." Individuals with scores below the mean tend to focus on the People style as their primary emphasis when deciding. They consider first the impact of choices and decisions on people's feelings and support, and on the need for harmony and positive relationships. Scores above the mean indicate a focus on the Task style. Those with this focus tend to look first at choices and decisions that are logical, sensible and can be justified objectively. They prefer making judgments that are impersonal, based on well-reasoned conclusions.

VIEW has demonstrated acceptable psychometric properties (Selby, Treffinger, & Isaksen, 2007a; Treffinger, 2005). The most recent master database for VIEW (as of the end of December, 2005) included 10,151 subjects. Based on 8,488 subjects who provided age data, the mean age is now 39.8 (SD = 11.6; range, 12 - 82). The database includes 4,316 male respondents (42.5%), 5,723 female respondents (56.4%), and 112 respondents (1.1%) who declined to state their gender. Table 1 summarizes several important descriptive statistics for each of VIEW's three dimensions: Orientation to Change (OC), Manner of Processing (MP), and ways of Deciding (WD), based on 10,151 responses.

The means for the 81 subjects included in this study were 67.59 for OC, 29.14 for MP and 38.96 for WD. This sample had a slightly more Exploratory preference for OC, was close to the observed mean for MP, and a slightly more Task oriented mean for WD.

For the purpose of this study, a survey was designed to assess the extent to which the participants enjoyed learning and used the CPS tools, guidelines and stages. The survey contained 33 questions for which the participants could indicate both their level of enjoyment of learning and use. A short summary of each of the 33 items was

included to help the participants remember to what item they were referring. The level of enjoyment was assessed through a question: "The extent to which I enjoyed

Table 1

VIEW Descriptive Statistics(N = 10,151)

Statistic	Orientation To Change	Manner Of Processing	Ways Of Deciding
Mean	74.6	30.1	34.6
Standard Deviation	15.8	9.2	8.5
Median	75.0	30.0	35.0
Mode	72.0	32.0	32.0
Minimum	18.0	8.0	8.0
Maximum	126.0	56.0	56.0
Skewness	-0.27	0.18	-0.15
Kurtosis	0.18	-0.27	-0.26
Standard Error of Measure	5.68	3.89	3.40

learning…" The five-point response scale included (1) Hated it, (2) Disliked it, (3) Neutral, (4) Liked it, and (5) Loved it. The level of use was assessed through the stem: "The extent to which I use…" The five-point response scale included (1) Never, (2) Rarely, (3) Sometimes, (4) Frequently, and (5) Very Often. The participants were also given the option to indicate that they did not know or did not remember. Parti-cipants were sent an explanation of the study as well as an invitation to participate and a link so that they could complete the survey on the web.

The 33 items provided the respondent the opportunity to make an assessment of both the level of enjoyment of their learning and their extent of use. The survey included items for 17 tools; eight were focusing tools like Highlighting, Paired Comparison Analysis, etc. Another nine items were for the generating tools like Visually Identifying Relationships, Brainwriting, etc. Eight additional items included the guidelines for both generating and focusing. The final eight items addressed the eight specific stages of CPS.

The design of the survey allowed the results to be collapsed into more meaningful categories. For example, the nine specific items relating to the generating tools could be combined for analysis, as could the focusing tools, generating and focusing guidelines, etc. The eight items relating to the stages of CPS could be collapsed to refer to the four main components.

RESULTS

The overall descriptive results for the survey are shown in Table 2. This table includees the mean, range and standard deviation for the extent of enjoyment and use for the collapsed categories from the survey. More detailed reporting of total item results are available elsewhere (Geuens, 2006; Isaksen & Geuens, 2006). For this sample, there are higher means for the extent to which the participants enjoyed learning over the extent to which they actually use CPS. Previous impact studies (Isaksen & Treffinger, 2004) have identified many possible barriers to the actual use of CPS including lack of opportunity, varying levels of management support and the climate or atmosphere within the workplace. These results are also consistent with participant feedback obtained immediately following the training courses.

Table 2
CPS Survey Descriptive Results (N = 81)

	Enjoy			Use		
	Range	Mean	SD	Range	Mean	SD
TOOLS						
Generating Tools	2.89-5.00	3.90	.42	1.00-4.67	2.98	.72
Focusing Tools	2.25-5.00	3.96	.48	1.00-4.75	3.37	.80
GUIDELINES						
Generating Guidelines	2.67-5.00	4.32	.52	1.00-5.00	4.00	.89
Focusing Guidelines	0.00-5.00	4.00	.75	0.00-5.00	3.66	.96
STAGES						
Understanding the Challenge	2.50-5.00	3.92	.66	1.00-5.00	3.50	.92
Generating Ideas	0.00-5.00	4.22	.88	0.00-5.00	3.93	1.23
Preparing for Action	0.00-5.00	3.70	1.12	0.00-5.00	3.26	1.24
Planning your Approach	0.00-5.00	3.73	1.31	0.00-5.00	3.56	1.41

Orientation to Change

To reduce the likelihood of a Type I error, the initial analysis included a multivariate analysis of variance (MANOVA) for scores on the Orientation to Change (OC) dimension of VIEW and the overall results on the survey. There was a significant interaction between OC and survey results, Wilk's Lambda = .01, $F(16,55) = 338.16$, $p < .0001$. The results for this dimension of VIEW are included in Table 3.

Explorers reported significantly higher levels of both enjoyment and use of Brainstorming with Post-its®. They also reported a significantly higher level of use of

Table 3
ANOVA Results for Orientations to Change (Df = 80)

	ENJOY				USE			
	Explorer (43)	Developer (38)	F	Sig	Explorer (43)	Developer (38)	F	Sig
TOOLS								
Generating Tools	3.96	3.84	1.81	.18	3.16	2.78	6.01	.02
Focusing Tools	3.99	3.92	.39	.54	3.55	3.18	4.50	.04
GUIDELINES								
Generating Guidelines	4.46	4.17	6.65	.01	4.28	3.69	10.05	.00
Focusing Guidelines	4.15	3.83	3.97	.05	3.89	3.41	5.32	.02
STAGES								
Understanding the Challenge	4.07	3.75	4.90	.03	3.75	3.22	7.22	.01
Generating Ideas	4.30	4.13	.76	.39	4.02	3.82	.57	.45
Preparing for Action	3.84	3.55	1.31	.26	3.55	2.93	5.17	.023
Planning your Approach	4.02	3.39	4.84	.03	3.94	3.12	7.389	.01

the Advantages, Limitations, Unique Qualities and Overcoming Limitations (ALUo) tool. Explorers also reported more enjoyment of two of the four guidelines for generating, and significantly higher use of all four of these guidelines. They also reported higher levels of enjoyment and use for considering novelty as one of the four guidelines for focusing, as well as higher levels of use for using affirmative judgment.

When it comes to the various stages within the current CPS framework, Explorers reported more enjoyment for learning both the Framing Problems and Generating Ideas stages.

After collapsing the items, Explorers reported significantly higher levels of enjoyment for the generating and focusing guidelines, and for Understanding the Challenge and Planning your Approach components. Explorers also reported significantly higher levels of use for both generating and focusing tools and guidelines, as well as all components other than Generating Ideas.

Manner of Processing

To reduce the likelihood of a Type I error, the initial analysis included a multivariate analysis of variance (MANOVA) for scores on the Manner of Processing (MP) dimension of VIEW and the overall results on the survey. There was a significant interaction between MP and survey results, Wilk's Lambda = .01, $F(16, 55) = 308.22$, $p < .0001$. The results for this dimension of VIEW are shown in Table 4. Internals reported significantly higher levels of enjoyment for learning the Brainwriting tool. Externals reported significantly higher levels of use of the Evaluation Matrix.

Externals reported higher levels of enjoyment for the generating guideline of striving for quantity. They also reported higher levels of enjoyment for the focusing guideline of using affirmative judgment. Externals also reported higher levels of use of the staying on course focusing guideline.

After collapsing the survey results into the more general categories, no significant differences were found between MP and the generating and focusing tools and guidelines or the CPS components.

Ways of Deciding

To reduce the likelihood of a Type I error, the initial analysis included a multivariate analysis of variance (MANOVA) for scores on the Ways of Deciding (WD) dimension of VIEW and the overall results on the survey. There was a significant interaction between WD and survey results, Wilk's Lambda = .01, $F(16, 55) = 328.87$, $p < .0001$.

The results for Ways of Deciding (WD) dimension of VIEW are shown in Table 5. All the significant differences on this dimension seemed to favor the Task-oriented deciders. They reported higher levels of use of the generating tools of Imagery Trek, Morphological Analysis, and Visually-Identifying Relationships. They also reported higher levels of use of Criteria.

Task-oriented deciders reported higher levels of enjoyment for learning to seek combinations when generating and to stay on course when focusing. They enjoyed the Constructing Opportunities stage of CPS and reported higher levels of use of Appraising Tasks.

After collapsing the results, Task-oriented deciders reported significantly higher enjoyment of learning the generating guidelines and the Generating Ideas component of CPS. Task-oriented deciders also reported significantly higher levels of use for generating and focusing tools, as well as the Generating Ideas and Planning your Appro-

ach components of CPS.

DISCUSSION AND CONCLUSIONS

It appears that there are meaningful style differences in the level of reported enjoy-

Table 4
ANOVA results for Manner of Processing (Df = 80)

	ENJOY				USE			
	External (45)	Internal (36)	F	Sig	External (43)	Internal (38)	F	Sig
TOOLS								
Generating Tools	3.93	3.87	.34	.56	3.03	2.92	.54	.47
Focusing Tools	3.99	3.92	.45	.51	3.50	3.21	2.71	.10
GUIDELINES								
Generating Guidelines	4.40	4.23	1.96	.17	4.08	3.91	.72	.40
Focusing Guidelines	4.04	3.95	.25	.62	3.77	3.53	1.29	.30
STAGES								
Understanding the Challenge	3.94	3.90	.06	.80	3.53	3.46	.10	.75
Generating Ideas	4.31	4.11	1.03	.31	4.02	3.81	.62	.44
Preparing for Action	3.68	3.74	.05	.82	3.46	3.01	2.59	.11
Planning your Approach	3.74	3.71	.02	.90	3.58	3.53	.03	.88

Table 5
ANOVA results for Way of Deciding (Df = 80)

	ENJOY				USE			
	People (39)	Task (42)	F	Sig	People (45)	Task (36)	F	Sig
TOOLS								
Generating Tools	3.88	3.93	.28	.60	2.82	3.14	4.13	.05
Focusing Tools	3.87	4.04	2.75	.10	3.17	3.56	5.20	.03
GUIDELINES								
Generating Guidelines	4.18	4.45	5.80	.02	3.85	4.14	2.13	.15
Focusing Guidelines	3.92	4.07	.79	.38	3.64	3.68	.04	.85
STAGES								
Understanding the Challenge	3.80	4.04	2.90	.09	3.29	3.70	4.07	.05
Generating Ideas	4.03	4.40	3.89	.05	3.64	4.19	4.18	.04
Preparing for Action	3.51	3.88	2.23	.14	3.03	3.48	2.73	.10
Planning your Approach	3.45	3.99	3.52	.06	3.22	3.87	4.47	.04

ment in learning CPS tools, guidelines and process. There are also differences in the reported level of use of these resources. This is consistent with other findings from the literature (Basadur, Graen & Wakabayashi, 1990; Basadur, Wakabayashi, & Graen, 1990; Basadur, Runco & Vega, 2000; Puccio, Wheeler & Cassandro, 2004; Runco & Basadur, 1993; Schoonover, 1996a &b).

Puccio, Wheeler and Cassandro (2004) found that students involved in CPS training were drawn to aspects of the course that were generally in alignment with their style preferences. Similar findings were reported by Basadur, Graen and Wakabayashi (1990); Basadur, Wakabayashi and Graen (1990); and Runco and Basadur (1993).

A challenge with these other findings is that the measures applied to examine style deliberately integrate specific models of creative problem solving. For example, Basadur and colleagues identify the conceptualizer, optimizer, implementor, and generator styles. These styles are inextricably linked to a sequential, yet closed circular model of CPS he calls Simplex (Basadur, 1991). Puccio and colleagues follow a remarkably similar approach. Their four styles include the clarifier, developer, ideator and implementor; reflecting preferences for problem identification, solution development, idea generation, and implementation, respectively. Both these approaches make it difficult to assess the extent to which the measures are truly general styles, or merely inventories of interest or preference for specific cognitive strategies or processes. Rather than landing within the clear and specific conceptual space of either person or process, these approaches appear to muddle the two. This situation makes it difficult to understand what exactly or precisely is being measured. The more specific question is: Are these measures of general cognitive or problem-solving styles or specific and more limited styles? Ultimately, the issue should be resolved by future research regarding the content and predictive validity of these measures.

An implication of this challenge creates yet another issue. Since these previous studies tie the measurement of process and person together, we must be especially mindful of the precise definition and validity of the model of process upon which the measures are based. The Simplex model is a derivative of one of the much earlier versions of the Osborn-Parnes CPS tradition. The Foursight (Puccio, 1999, 2002) assessment is based on a later version of CPS, but not the current one (Isaksen & Treffinger, 2004; Treffinger & Isaksen, 2005). The model of CPS used in this current study is CPS version 6.1 that includes not only the tools, guidelines, language and process stages, but also meta-cognitive or process planning stages (Treffinger, Isaksen & Stead-Dorval, 2006). As a result, it is difficult to make direct comparisons based on previous (and future) research into stylistic preferences for learning and using CPS. Although deploying essentially a similar set of tools and guidelines, the names or titles of these items have changed from their original source.

This challenge and related issue, make it difficult for the emerging field of creativity and innovation to adequately deal with the required analytic simplification and synthetic coordination to move the field forward (Isaksen, Murdock, Firestien, & Treffinger, 1993). This is particularly true for inquiry designed to better understand how to best stimulate creativity (Treffinger, 1993). Further research must be conducted using a common vocabulary and conceptual schema.

VIEW was designed to specifically assess problem-solving style as a general concept, firmly and clearly grounded within the domain of person. The VIEW theory and measure are based on understanding three independent dimensions of style. Most of

our previous research within the CSP utilized the KAI, which does correlate with the OC dimension of VIEW. Based on this study, the two additional dimensions of VIEW did add more value in identifying and understanding other style differences related to learning and applying CPS. Some evidence regarding VIEW's impact on learning about creativity and CPS in the business context (Esposito & Roehm, 2004) and the educational context (Treffinger, 2006) is already available. Given the conceptual clarity of VIEW, and independence from a specific model of process, future research might also be helpful in understanding not only CPS, but also other change methods (Isaksen & Tidd, 2006). This research could also include a consideration of compareing results from educational and business contexts (see Treffinger, Selby & Isaksen, In press).

This was an exploratory study designed to better understand style differences in learning and using CPS. As such, there is a great deal more research that is needed before any definitive comments and conclusions are drawn. First, the sample was compiled from a series of ICP courses given over the past two years. The reported range of the time between the actual training and completing the assessment ranged from five months to 16 years, with a mean of 21 months. One participant had reported his first exposure to training 16 years ago, even though he had attended a much more recent ICP course.

Other studies have examined participants' reactions to the training immediately upon completion (i.e. Puccio, Wheeler, & Cassandro, 2004). As a result, it would be important to explore the issue of the interval between the training and learning and the actual completion of the survey. If creativity and innovation practitioners are interested in the full spectrum of impact of training (Kirkpatrick, 1996), then this issue should be addressed in future research. This may help us better understand how long it takes to acquire higher level of impact and any differences of time after training. For example, style differences in both enjoyment and use could be affected by how long participants have had to apply their learning after the initial training.

The current design for the ICP course is three days. Earlier versions were four or five days. It would also be interesting to determine the most productive design in terms of both enjoyment and actual use of CPS. The current three-day design includes a short tutorial available through the Internet. It would be interesting to better understand the nature of the delivery system and its relationship to both style preferences and impact. What would be the most effective design and delivery of the training?

Since the data was acquired via survey, we relied on self-report of both enjoyment and use. It would be better to examine actual behavior rather than rely on self-reports. This could include examining some more objective observations and evaluations of actual performance or results indicators, as well as assessments of the kind of task and climate within which the performance was observed. The survey design may need to be reconsidered so that it includes more narrative input as well. This kind of research would lead us toward a much richer ecological program of research.

Given the exploratory nature of this study, there are numerous limitations that must be identified. One is the size and kind of sample. Although there was a good spread within each of the dimensions of VIEW, a sample size of 81 precludes any definitive conclusions. The sample means showed that this group was more Exploratory, External, and Task oriented. This kind of research must include larger samples that reflect the means derived on VIEW. In addition, this study did not examine the potential influence of gender, age, time in the current work situation, etc. These factors should be

topics for further research.

Further, it is not certain what kind of impact learning about participants' styles may have had on their self-report assessment of enjoyment and use of CPS. It may be useful to find ways to examine style differences when the participants are not consciously aware of their styles to avoid response bias.

Given the limitations to this exploratory study, there is a clear indication that problem-solving style, as measured by VIEW, does interact with the learning and application of CPS tools, guidelines and process. VIEW is a productive tool to help participants understand their natural preferences for certain aspects of CPS and further research can assist those who are engaged in both learning and applying this framework.

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